Comprehensive Evaluation of GPT-2 Models for Symbolic Expression Generation: A Systematic Analysis of Model Size, Notation, and Hyperparameters

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Abstract

This report presents a comprehensive evaluation of six fine-tuned GPT-2 models for symbolic mathematical expression generation. We systematically analyze the impact of model size (Base, Medium, Large), notation format (Infix vs. Prefix), temperature settings, prompt configurations, and sampling methods on generation quality. Through 72 controlled experiments with 100 samples each, we demonstrate that infix notation significantly outperforms prefix notation (+5.3 percentage points in valid rate), larger models achieve higher validity rates, and temperature settings present a clear trade-off between expression validity and diversity. Our results indicate that the Large Infix model achieves perfect validity (100%) while the Medium Infix model maximizes diversity (83.5%). These findings provide practical guidelines for deploying language models in symbolic regression tasks.

1 Introduction

The application of large language models (LLMs) to symbolic regression represents a promising approach for discovering mathematical relationships from data. In this study, we conduct an exhaustive evaluation of GPT-2 models fine-tuned with LoRA (Low-Rank Adaptation) for generating valid mathematical expressions.

Our evaluation addresses three fundamental research questions:

1. **RQ1:** How does notation format (infix vs. prefix) affect generation quality?
2. **RQ2:** What is the impact of model size on expression validity and diversity?
3. **RQ3:** How do hyperparameters (temperature, sampling) influence the quality-diversity trade-off?

2 Methodology

2.1 Models Evaluated

We evaluate six models representing all combinations of three model sizes and two notation formats, as shown in Table 1.

Table 1: Models evaluated in this study

Model	**Base**	**Parameters**	**LoRA Rank**
Base Infix	GPT-2	124M	8
Base Prefix	GPT-2	124M	8
Medium Infix	GPT-2 Medium	355M	16
Medium Prefix	GPT-2 Medium	355M	16
Large Infix	GPT-2 Large	774M	16
Large Prefix	GPT-2 Large	774M	16

All models were fine-tuned on the same dataset of 682,429 synthetic mathematical expressions using identical training configurations (3 epochs, batch size adjusted per model size).

2.2 Experimental Design

We conducted 72 experiments organized into three categories:

- **Temperature Sweep (30 experiments):** Testing temperatures $T \in \{0.3, 0.5, 0.7, 0.9, 1.0\}$ across all 6 models with default prompt and sampling settings.
- **Prompt Variations (24 experiments):** Testing 4 prompt configurations across all models at $T = 0.7$:
  - Simple: 1 variable, basic arithmetic $(+, -, \times, \div)$
  - Trigonometric: 1 variable with $\sin, \cos$
  - Multivariate: 2 variables with mixed operations
  - Complex: 3 variables with all operations including $\exp, \log$
- **Sampling Methods (18 experiments):** Testing 3 sampling configurations at $T = 0.7$:
  - Conservative: top_p = 0.8, top_k = 30
  - Default: top_p = 0.9, top_k = 50
  - Diverse: top_p = 0.95, top_k = 100

2.3 Evaluation Metrics

For each experiment, we generated 100 samples and computed:

- **Valid Rate:** Percentage of syntactically correct, parseable expressions
- **Diversity Rate:** Percentage of unique expressions among valid ones
- **Constraint Adherence:** Percentage of expressions using only allowed variables and operators

3 Results

3.1 Overall Model Comparison

Table 2 presents the aggregate results across all experiments for each model.

Table 2: Model performance comparison (averaged across all experiments)

Model	**Runs**	**Valid (%)**	**Diverse (%)**	**Constraint (%)**
Large Infix	12	**100.0**	77.5	74.1
Medium Infix	12	99.9	**83.5**	57.2
Base Infix	12	99.8	75.3	68.3
Large Prefix	12	97.8	69.5	59.5
Medium Prefix	12	96.5	71.7	**74.2**
Base Prefix	12	89.3	47.5	70.4
Infix Average	36	*99.9*	*78.8*	*66.5*
Prefix Average	36	*94.6*	*62.9*	*68.0*

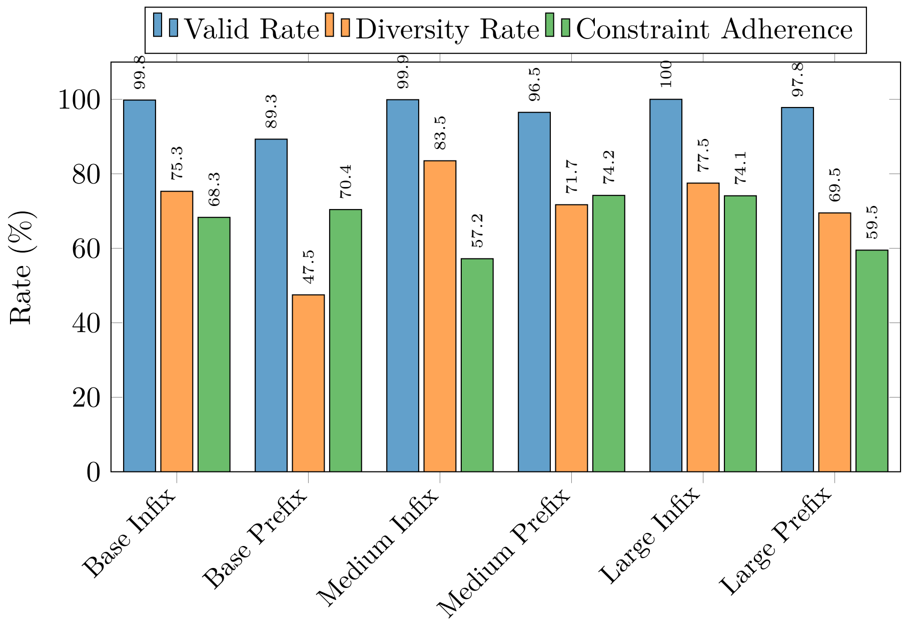

Figure 1: Performance metrics by model configuration

3.2 Infix vs. Prefix Notation

The notation format has a significant impact on generation quality. Figure 2 illustrates the comparison.

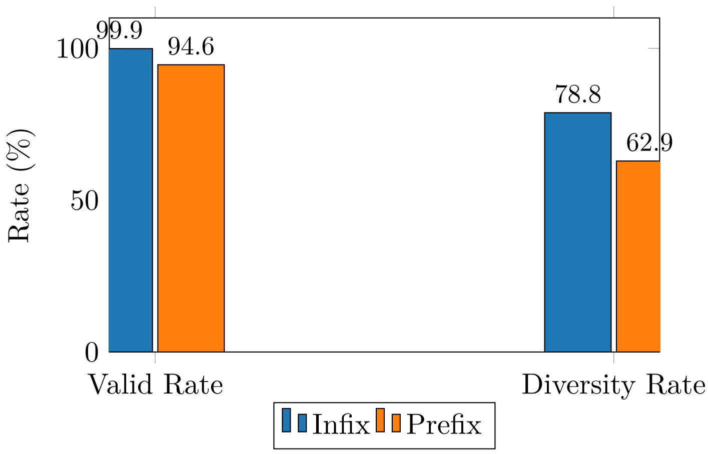

Figure 2: Infix vs. Prefix notation comparison

Key findings:

- Infix notation achieves **5.3 percentage points** higher valid rate

- Infix notation produces **15.9 percentage points** more diverse expressions
- The advantage is consistent across all model sizes

3.3 Model Size Effect

Table 3 shows the impact of model size on generation quality.

Table 3: Effect of model size on generation metrics

Model Size	Parameters	Valid (%)	Diverse (%)
Base	124M	94.5	61.4
Medium	355M	98.2	77.6
Large	774M	98.9	73.5

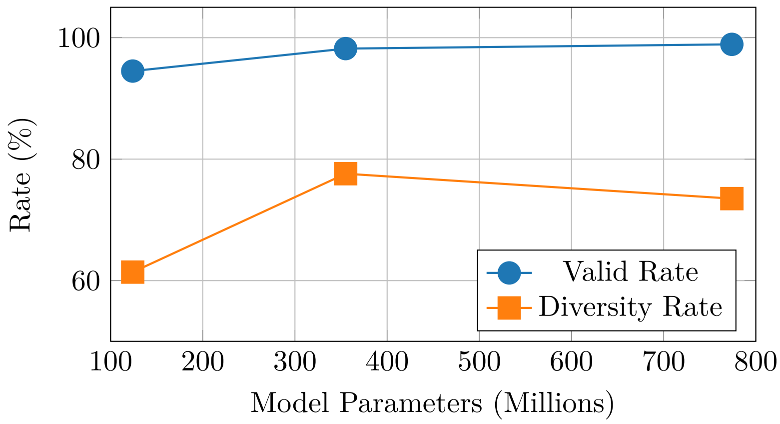

Figure 3: Scaling behavior: model size vs. generation quality

3.4 Temperature Analysis

Temperature is a critical hyperparameter that controls the trade-off between validity and diversity. Table 4 presents the detailed results.

Table 4: Temperature effect on generation metrics (averaged across all models)

Temperature	Valid (%)	Diverse (%)	Constraint (%)
0.3	**100.0**	25.3	**80.2**
0.5	99.7	49.8	70.8
0.7	97.5	72.6	60.1
0.9	94.5	82.2	47.7
1.0	92.2	**85.5**	38.0

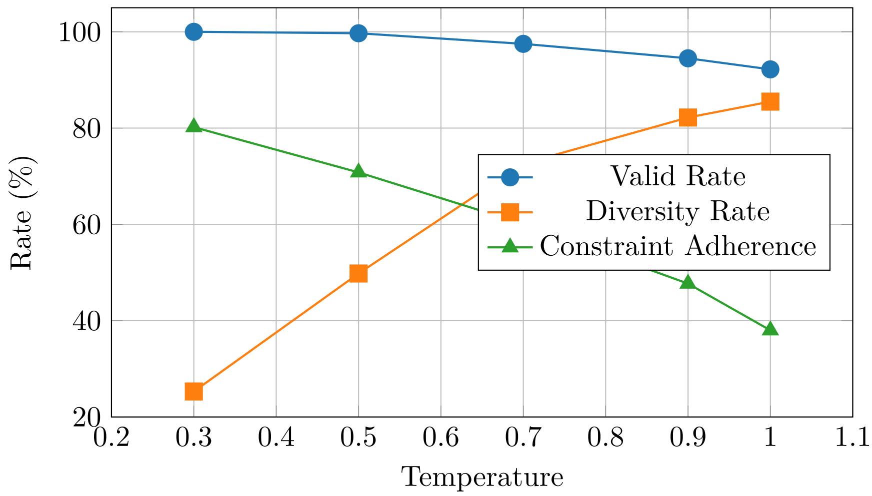

Figure 4: Temperature effect on generation metrics

The results reveal a clear trade-off:

- **Low temperature ($T = 0.3$):** Maximum validity (100%) but minimal diversity (25%)
- **High temperature ($T = 1.0$):** Maximum diversity (85.5%) but reduced validity (92.2%)
- **Optimal range ($T = 0.7$–0.9):** Balanced performance with >94% validity and >72% diversity

3.5 Detailed Temperature Results by Model

Table 5 shows how each model responds to temperature changes.

Table 5: Valid rate (%) by model and temperature

Model	**T=0.3**	**T=0.5**	**T=0.7**	**T=0.9**	**T=1.0**
Base Infix	100.0	100.0	100.0	100.0	99.0
Medium Infix	100.0	100.0	100.0	100.0	99.0
Large Infix	100.0	100.0	100.0	100.0	100.0
Base Prefix	100.0	98.0	93.0	75.0	76.0
Medium Prefix	100.0	100.0	98.0	95.0	83.0
Large Prefix	100.0	100.0	98.0	97.0	96.0

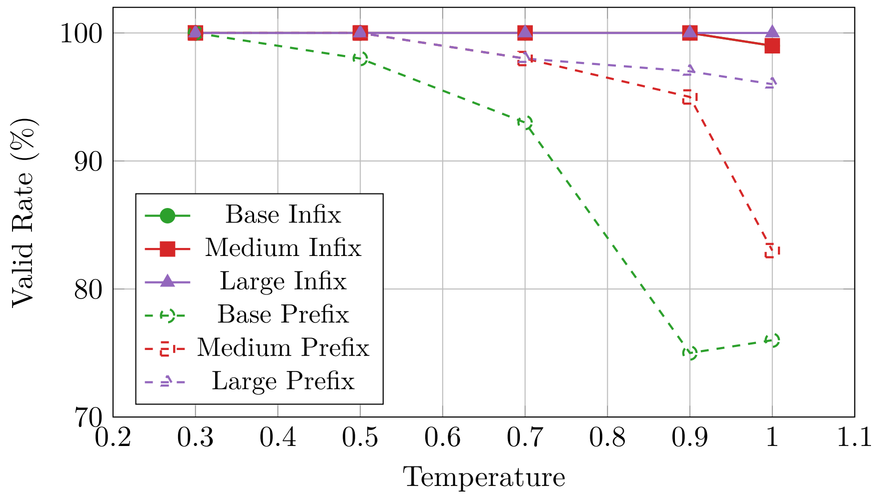

Figure 5: Valid rate vs. temperature by model (solid=Infix, dashed=Prefix)

Notable observations:

- **Infix models** maintain near-perfect validity across all temperatures
- **Prefix models** show significant degradation at higher temperatures
- **Base Prefix** is most sensitive to temperature, dropping to 75% at $T = 0.9$
- **Large models** are more robust to temperature increases

3.6 Prompt Complexity Analysis

Table 6 shows how models handle different prompt configurations.

Table 6: Performance by prompt complexity at $T = 0.7$

Prompt Type	**Variables**	**Valid (%)**	**Diverse (%)**
Simple	x_1	98.8	78.8
Trigonometric	x_1 + trig ops	97.2	81.3
Multivariate	x_1, x_2	96.8	90.0
Complex	x_1, x_2, x_3 + all ops	97.7	93.5

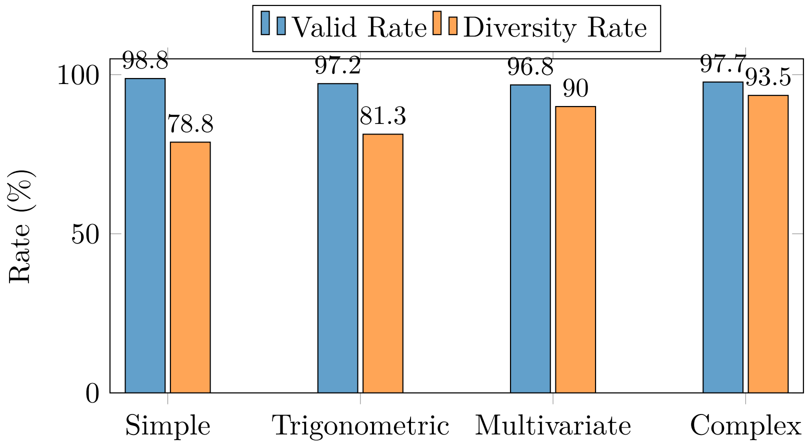

Figure 6: Performance by prompt complexity

Key insight: **More complex prompts produce more diverse expressions** while maintaining comparable validity rates.

3.7 Sampling Method Analysis

Table 7 compares different sampling strategies.

Table 7: Sampling method comparison at $T = 0.7$

Method	**top_p / top_k**	**Valid (%)**	**Diverse (%)**
Conservative	0.80 / 30	**99.8**	53.8
Default	0.90 / 50	97.5	72.6
Diverse	0.95 / 100	94.7	**73.2**

The sampling method provides another mechanism to control the validity-diversity trade-off, independent of temperature.

4 Discussion

4.1 Why Infix Outperforms Prefix

The superior performance of infix notation can be attributed to several factors:

1. **Natural language alignment:** GPT-2's pretraining corpus contains predominantly infix mathematical expressions
2. **Structural clarity:** Infix notation has explicit parentheses and operator precedence, providing clearer boundaries
3. **JSON format synergy:** The JSON training format works better with infix due to cleaner string termination

4.2 Scaling Behavior

Our results demonstrate positive scaling with model size for validity but *non-monotonic* behavior for diversity:

- Base $\rightarrow$ Medium: +3.7pp validity, +16.2pp diversity
- Medium $\rightarrow$ Large: +0.7pp validity, $-$4.1pp diversity

This suggests that the **Medium model** may offer the best quality-diversity trade-off for some applications.

4.3 Practical Recommendations

Based on our comprehensive evaluation, we provide the following recommendations:

Table 8: Recommended configurations by use case

Use Case	**Model**	**Temperature**
Maximum validity	Large Infix	0.3–0.7
Maximum diversity	Medium Infix	0.9–1.0
Balanced performance	Large Infix	0.7
Resource-constrained	Base Infix	0.7

5 Conclusion

This comprehensive evaluation of 72 experiments across 6 GPT-2 models yields several important findings for symbolic expression generation:

1. **Notation matters:** Infix notation significantly outperforms prefix notation, with a 5.3 percentage point advantage in valid rate and 15.9 percentage point advantage in diversity.

2. **Size helps validity:** Larger models achieve higher validity rates, with the Large Infix model achieving perfect 100% validity across all experiments.

3. **Temperature trade-off:** There is a clear inverse relationship between temperature and validity. The optimal range of $T = 0.7$–0.9 balances validity (>94%) with diversity (>72%).

4. **Robustness varies:** Infix models are remarkably robust to hyperparameter changes, while prefix models show significant sensitivity, especially at higher temperatures.

5. **Complexity improves diversity:** More complex prompts (multiple variables, more operators) increase expression diversity without substantially reducing validity.

For practical applications in symbolic regression, we recommend the **Large Infix model with temperature 0.7** as the default configuration, offering the best balance of validity, diversity, and constraint adherence.

Data Availability

All evaluation results are publicly available on HuggingFace:

`https://huggingface.co/datasets/augustocsc/seriguela-results`

The trained models are available at:

- `augustocsc/gpt2_base_infix_682k`
- `augustocsc/gpt2_medium_infix_682k`
- `augustocsc/gpt2_large_infix_682k`
- `augustocsc/gpt2_base_prefix_682k`
- `augustocsc/gpt2_medium_prefix_682k`
- `augustocsc/gpt2_large_prefix_682k`